

Comparative Analysis of Depreciation Methods: Straight- Line vs. Diminishing Balance

A Detailed Report on the Financial Implications of Depreciation Methods

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INTRODUCTION TO DEPRECIATION



Depreciation is the systematic allocation of the cost of a tangible asset over its useful life.



It reflects the reduction in value of an asset as it is used over time.



Two common methods of depreciation are:

Straight-Line Method

Diminishing Balance Method



This report compares these two methods and provides practical examples.

OBJECTIVES OF THE PRESENTATION



1. To explain the Straight-Line and Diminishing Balance depreciation methods.



2. To compare their financial implications.



3. To demonstrate calculations and examples for both methods.



4. To analyze their suitability for different assets.

KEY TERMINOLOGIES

- Asset Cost: Initial purchase price of the asset.
- Additional Asset Cost: Extra costs (e.g., transportation, installation).
- Asset Price: Total cost (Asset Cost + Additional Costs).
- Scrap Value: Residual value at the end of the asset's life.
- Estimated Life Span: Duration (in years) the asset will be used.
- Depreciation/Year: Annual depreciation expense.
- Depreciation Percentage: Annual depreciation rate.

Straight-Line Method

Formula:

$\text{Depreciation per year} = (\text{Asset Price} - \text{Scrap Value}) / \text{Estimated Life Span}$

Example:

- Asset Price = \$500,000

- Scrap Value = \$50,000

- Life Span = 10 years

- Depreciation/Year = $(\$500,000 - \$50,000) / 10 = \$45,000$

- Depreciation Schedule:

- - Year 1: \$45,000 (Book Value = \$455,000)

- - Year 2: \$45,000 (Book Value = \$410,000)

- - ...

- - Year 10: \$45,000 (Book Value = \$50,000)

Diminishing Balance Method

Formula:

$\text{Depreciation per year} = \text{Book Value at Beginning of Year} * \text{Depreciation Rate}$

Example:

-
- Asset Price = \$500,000
-
- Scrap Value = \$50,000
-
- Life Span = 10 years
-
- Depreciation Rate ~ 20.6%
-

- Depreciation Schedule:

- - Year 1: $\$500,000 * 20.6\% = \$103,000$ (Book Value = $\$397,000$)

- - Year 2: $\$397,000 * 20.6\% = \$81,782$ (Book Value = $\$315,218$)

- - ...

- - Year 10: Depreciation adjusts to reach Book Value ~ $\$50,000$

Aspect	Straight-Line Method	Diminishing Balance Method
Annual Depreciation	Fixed (\$45,000/year)	Higher early, lower later
Simplicity	Easy to calculate	More complex
Suitability	Consistent utility assets	Assets losing value quickly
Total Depreciation	\$450,000	\$450,000

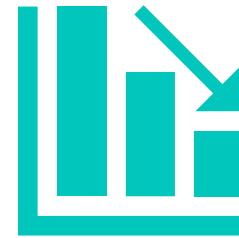
Comparative Analysis

Straight-Line Method



Advantages:

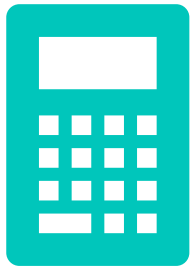
Simple to calculate.
Consistent annual expense.



Disadvantages:

Does not reflect accelerated loss
of value.

Diminishing Balance Method:



Advantages:

Reflects actual usage and value loss.



Disadvantages:

More complex calculations.
Lower book values in later years.

Problem Statements

- Straight-Line Method:
 1. Calculate Asset Price.
 2. Calculate Depreciation/Year.
 3. Calculate Total Depreciation.
 4. Find Book Value after Life Span.

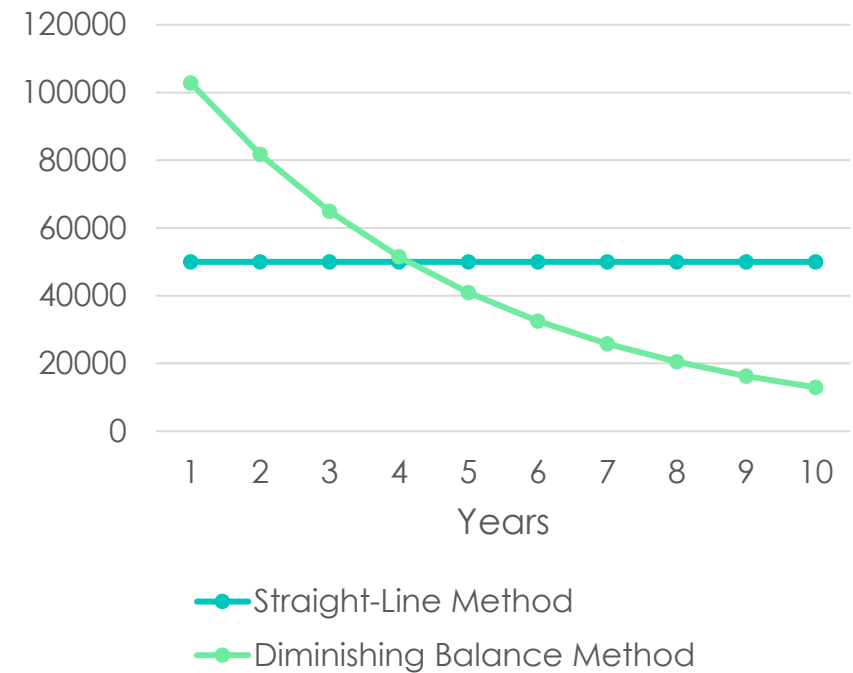
- Diminishing Balance Method:
 1. Calculate Asset Price.
 2. Determine Depreciation Rate.
 3. Calculate Year-on-Year Depreciation and Book Value.

Advanced Problem Solutions

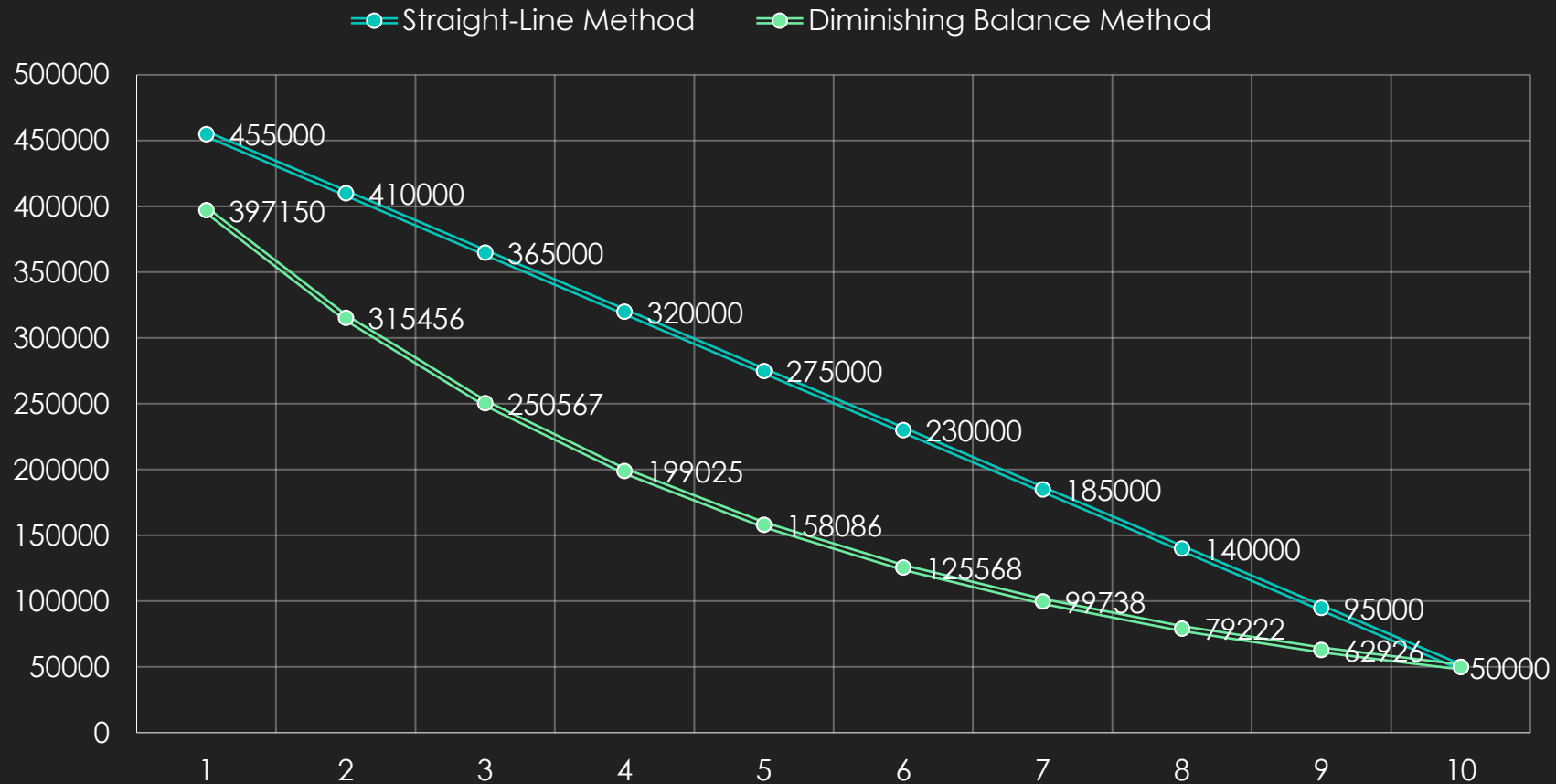
- ❖ 1. Straight-Line Annual Depreciation: \$45,000 ($=\text{SLN}(\text{D8},\text{D9},\text{D10})$)
- ❖ 2. Straight-Line Total Depreciation: \$450,000 ($= \text{D11}*\text{D10}$)
- ❖ 3. Book Value After Life (Straight-Line): \$50,000 ($=\text{D8}-\text{D13}$)
- ❖ 4. Diminishing Balance Rate: 20.6% ($=1-(\text{D21}/\text{D20})^{(1/\text{D22})}$)
- ❖ 5. Year 2 Depreciation (Diminishing): \$81,782 ($=\text{IF}(\text{D27}>\$D\$21,(\text{D27}*\$D\$23))$)
- ❖ 6. Year 4 Book Value (Diminishing): \$2,50,593.51 ($=\text{D28}-\text{C28}$)
- ❖ 7. Total Depreciation (Diminishing): \$450,000 ($=\text{D20}-\text{D21}$)
- ❖ 8. Book Value After Life (Diminishing): \$50,000 ($=\text{D21}$)
- ❖ 9. Which Method Results in Higher Total Depreciation? Both the **Straight-Line Method** and the **Diminishing Balance Method** result in the **same total depreciation** of **\$450,000**. Neither method results in higher total depreciation.

Visual Representation

Annual Depreciation
(Straight-Line vs. Diminishing Balance)



BOOK VALUE TRENDS OVER 10 YEARS (STRAIGHT-LINE VS. DIMINISHING BALANCE)



Conclusion

- Both methods suit different needs:
- Straight-Line: Best for assets with steady utility.
- Diminishing Balance: Ideal for assets losing value quickly.
- The choice depends on asset type and financial goals.

Tools and Technologies Used

- PowerPoint: For creating this presentation and visuals.
- Excel: For calculations and depreciation schedules.
- Graphs: Created using Excel for visual representation.